

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458597.8492 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1374 +/- 0.005
Transit depth (Rp/Rs)^2	1.89 +/- 0.14 [%]
Orbital Inclination (inc)	89.95 +/- 0.37
Airmass coefficient 1 (a1)	1.2294 +/- 0.0084
Airmass coefficient 2 (a2)	-0.201 +/- 0.019
Scatter in the residuals of the lightcurve fit is	0.69 %
Best Comparison Star	None
Optimal Aperture	4.91
Optimal Annulus	12.74
Transit Duration (day)	0.12692 +/- 0.00067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1374 ± 0.005 vs 0.1326 (deviation 0.79σ)
Duration fit vs published	0.12692 d vs 0.1238 d (deviation 3.92σ)
Mid-time O–C	4.4 min
Depth SNR	22.8
Residual scatter	0.69 %

Light curve

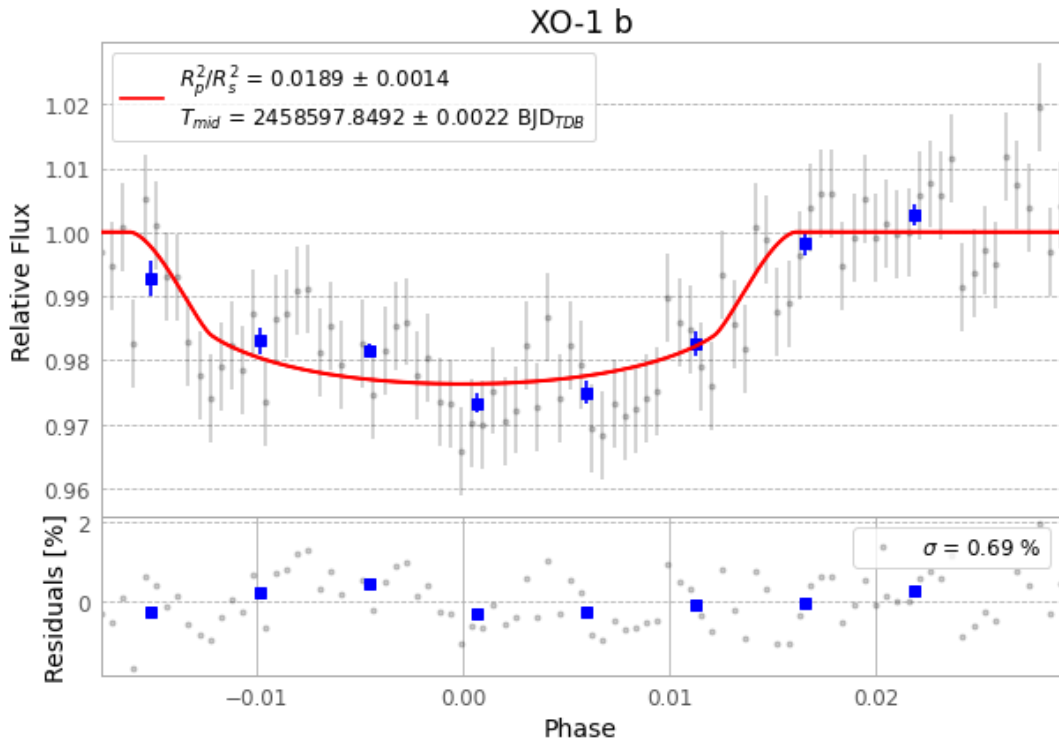


Figure 1 — FinalLightCurve_XO-1 b_2019-04-23.png (SHA-256 f143543bc4d5031e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [396, 122], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b0f7242a25a96235...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

89 FITS frames (58 MB), XO-1190424063612.FITS → XO-1190424110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-190424.log (a043dedcd9f6...), FinalLightCurve_XO-1 b_2019-04-23.png (f143543bc4d5...), FinalLightCurve_XO-1 b_2019-04-23.csv (a235b90dda0a...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 06:19Z.